A Proof of Detlefs' Harmonic-Number Primality Criterion on OEIS A000040

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Abstract
In September 2010, G. Detlefs contributed to OEIS A000040 a pair of conjectural primality criteria. The first, that the primes are the $n \neq 4$ with $n! \equiv n(n-1) \pmod{n^2}$, was settled on the entry in May 2026. The second, that they are the $n \neq 4$ with $n!\,H_n \equiv n-1 \pmod{n}$ where $H_n = \sum_{k=1}^{n} 1/k$, still stands unproved. We prove it. The argument is four lines: in the integer $n!\,H_n = \sum_{k=1}^{n} n!/k$ every summand with $k < n$ is a multiple of n, so $n!\,H_n \equiv (n-1)! \pmod{n}$ and the criterion is Wilson's theorem verbatim. We note in addition that the stated hypothesis $n \neq 4$ is doing no work in this second criterion — unlike in the first, $n = 4$ fails the congruence of its own accord — so the criterion holds for every $n \geq 2$ without exception.

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1 The conjecture

On 10 September 2010, Gary Detlefs contributed to OEIS A000040, the sequence of primes, the following pair of statements under a single heading [1]:

> *Conjecture:*
>
> $a(n) = \{n \mid n! \bmod n^2 = n(n-1)\}$, $n \neq 4$.
>
> $a(n) = \{n \mid n!\,h(n) \bmod n = n-1\}$, $n \neq 4$, *where* $h(n) = \sum_{k=1}^{n} 1/k$.

The first of the two now carries an editorial annotation on the entry recording that it follows from Wilson's theorem (R. Ahmed, 21 May 2026). The second carries no such annotation and remains, as of 22 August 2026, an unproved conjecture on the live entry; the page carried the timestamp 22 August 2026.

Write $H_n = h(n) = \sum_{k=1}^{n} 1/k$. The quantity $n!\,H_n$ is an integer, since

$$n!\,H_n \;=\; \sum_{k=1}^{n} \frac{n!}{k}, \tag{1}$$

and each $n!/k$ with $1 \leq k \leq n$ is an integer. This is OEIS A000254 up to indexing, the unsigned Stirling numbers of the first kind $c(n+1, 2)$. The assertion to be proved is:

Theorem 1. *For every integer $n \geq 2$,*

$$n!\,H_n \equiv n-1 \pmod{n} \iff n \text{ is prime.}$$

2 Proof

Lemma 2. *For every integer $n \geq 2$,*

$$n!\, H_n \;\equiv\; (n-1)! \pmod{n}.$$

Proof. Use (1). Fix k with $1 \leq k \leq n-1$. Then $k \leq n-1$, so k divides $(n-1)!$, and therefore

$$\frac{n!}{k} \;=\; n \cdot \frac{(n-1)!}{k}$$

is n times an integer; hence $n!/k \equiv 0 \pmod{n}$. The single remaining summand is the one with $k = n$, namely $n!/n = (n-1)!$. Summing, $n!\, H_n \equiv (n-1)! \pmod{n}$. □

Proof of Theorem 1. By Lemma 2 the congruence $n!\, H_n \equiv n-1 \pmod{n}$ is equivalent to

$$(n-1)! \equiv n-1 \equiv -1 \pmod{n},$$

which by Wilson's theorem together with its converse holds precisely when n is prime. (For completeness: if $n = p$ is prime, pairing each of $2, \ldots, p-2$ with its inverse modulo p leaves $1 \cdot (p-1) \equiv -1$. If n is composite and $n \neq 4$, write $n = ab$ with $1 < a < b < n$, or $n = p^2$ with $p \geq 3$ and then $p, 2p \leq n-1$ are distinct; in either case $n \mid (n-1)!$, so $(n-1)! \equiv 0 \not\equiv -1$, the last step using $n > 1$. For $n = 4$, $(4-1)! = 6 \equiv 2 \not\equiv -1 \pmod{4}$.) □

Remark 3 (the hypothesis $n \neq 4$ is idle here). Detlefs attaches the caveat $n \neq 4$ to both criteria. In the second it is unnecessary: $4!\, H_4 = 24+12+8+6 = 50$ and $50 \equiv 2 \pmod{4}$, while $n - 1 = 3$, so $n = 4$ fails the congruence on its own and needs no excluding. The caveat is a carry-over from the first criterion, where the modulus is n^2 and the exceptional behaviour of 4 is a genuine feature. Theorem 1 is therefore stated for all $n \geq 2$ with no exceptions.

Remark 4 (what the harmonic number contributes). Nothing. Lemma 2 shows that modulo n the entire sum $\sum_{k<n} n!/k$ vanishes, so the criterion sees only the final term $(n-1)!$. The same collapse happens for any weighted sum $\sum_{k=1}^{n} c_k\, n!/k$ with $c_n = 1$: the harmonic weights play no role beyond making the $k = n$ coefficient equal to 1. In this sense the criterion is Wilson's theorem in disguise, and the disguise is thin.

Corollary 5. *For $n \geq 2$, n is prime if and only if n divides $n!\, H_n + 1$.*

Proof. $n!\, H_n \equiv n - 1 \equiv -1 \pmod{n}$ is the same as $n \mid n!\, H_n + 1$. □

3 Computational verification

The result is proved, so computation is a guard against error in the statement rather than evidence. A short script (Python 3, standard library plus `sympy`) checks the following; all pass.

1. For $2 \leq n < 600$, computing the exact integer $n!\,H_n = \sum_{k=1}^{n} n!/k$ and reducing modulo n, the predicate "$n!\,H_n \bmod n = n-1$" agrees with "n is prime" in every case, with no exceptions.

2. For $2 \leq n < 400$, $n!\,H_n \equiv (n-1)! \pmod{n}$, confirming Lemma 2.

3. $4!\,H_4 = 50$ and $50 \bmod 4 = 2 \neq 3$, confirming that $n = 4$ is excluded automatically.

Every numeral in this note is an output of that script.

4 Concluding remarks

The proof is elementary and short; we claim no difficulty for it. It is recorded because the statement has stood on A000040 as an unproved conjecture since September 2010, because its companion — posted in the same comment on the same day — was annotated as settled in May 2026 while this one was left untouched, and because the reduction in Lemma 2 shows the criterion to be a restatement of Wilson's theorem rather than a new test. The natural destination is a comment on the entry rather than a journal.

We have found no earlier proof on the entry or in the literature, but an argument this short could well have been given before, and we would welcome notice of any earlier appearance.

References

[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`, 2026. Sequence A000040 (the primes); comment of G. Detlefs, 10 September 2010, and annotation of R. Ahmed, 21 May 2026. Cf. A000254.